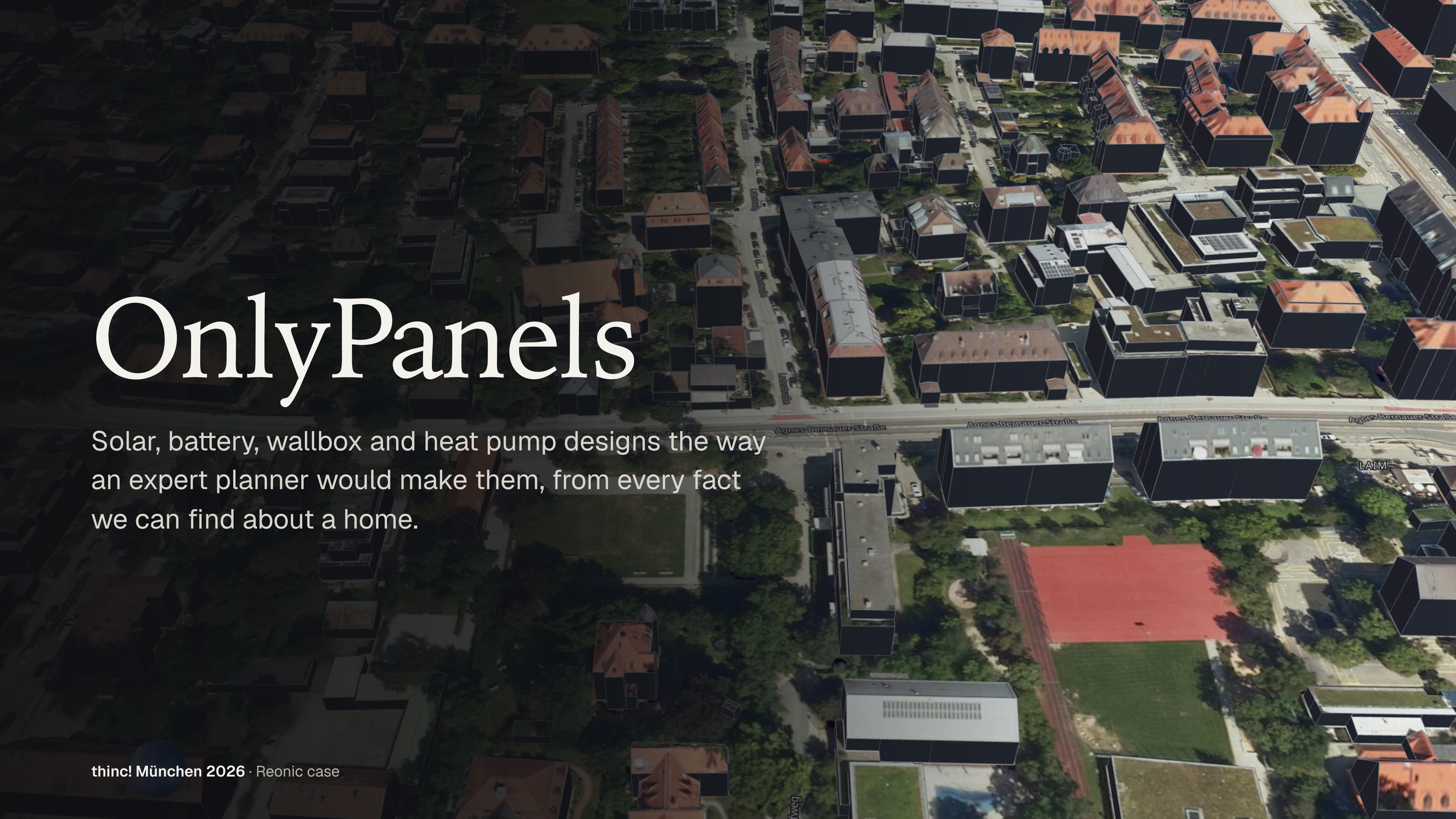


OnlyPanels

Solar, battery, wallbox and heat pump designs the way an expert planner would make them, from every fact we can find about a home.

thinc! München 2026 · Reonic case

Team



Atahan Kılıç

?

Utku Kalkanlı

Same request. Different offer.

59%

of Reonic's solar requests match another customer's, field for field

99%

of those identical requests still got a different design from the expert

6.3 kWp

apart on average: a whole roof side

The experts knew what the request never said: the roof, the shade, the street. AI is smart enough.
What it lacks is context.

1,306 solar projects, 2 installers · scripts/profile_same_request.py

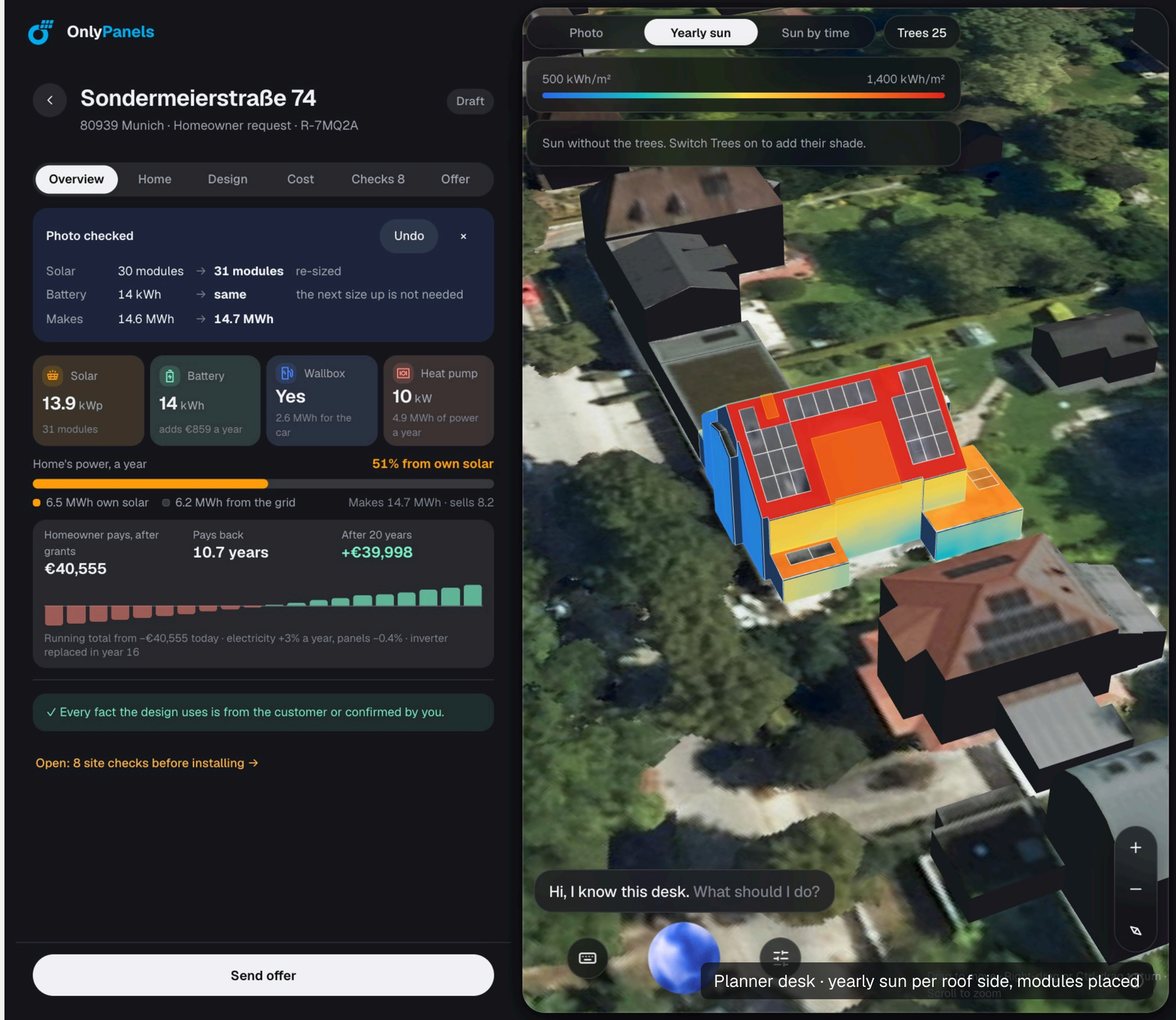
One click on a house. Twelve sources, one draft.

1 **Measured:** 3D roof, aerial photo, sun per roof side, tree shade

2 **Found:** the neighbourhood's age and heating, district heating plans, monument rules

3 **Priced:** the installer's own price book, grants and energy law

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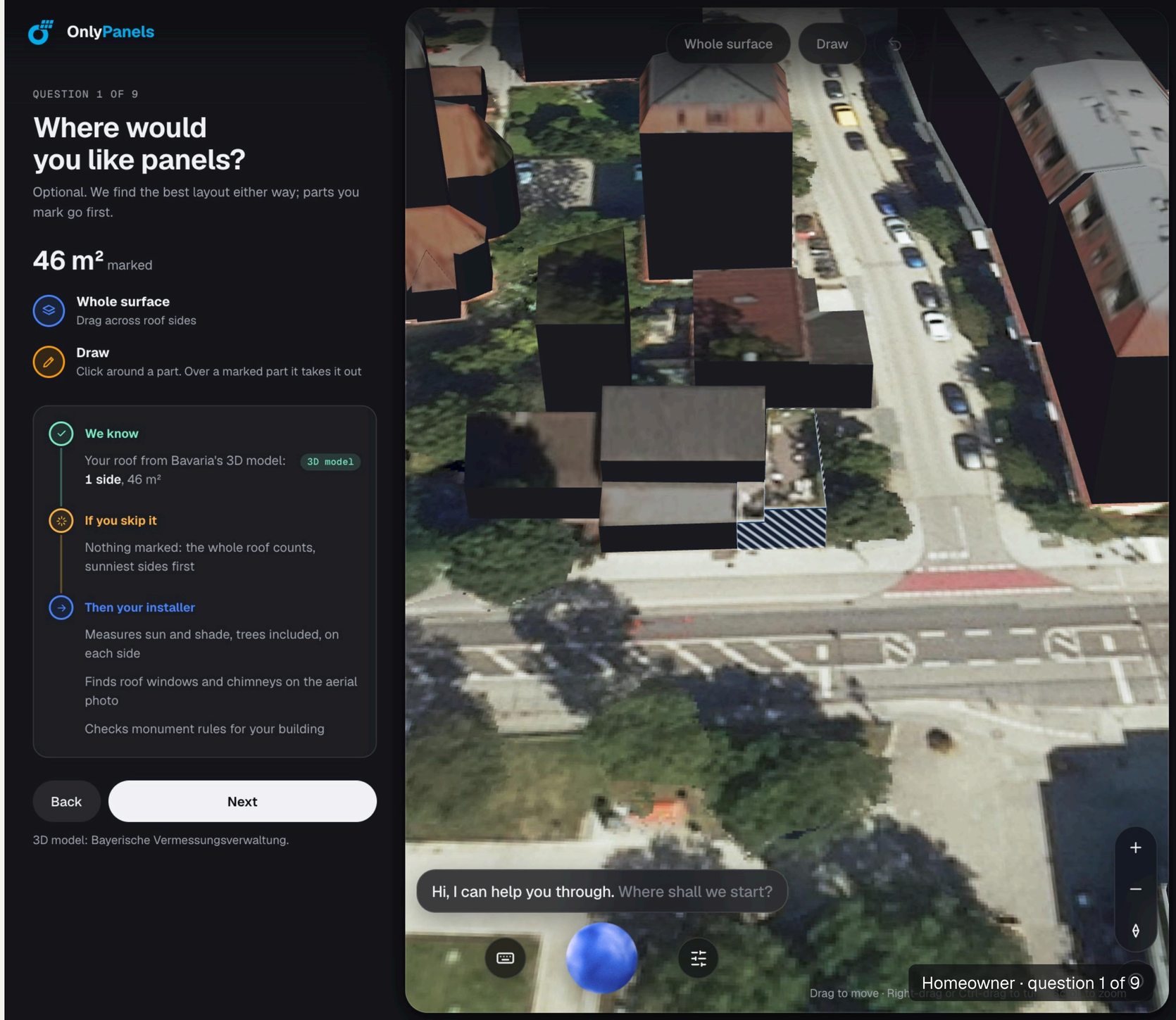
Your home stays in your browser.

1. **The browser fetches the facts itself,** from Bavarian and EU services. Nothing about the home reaches us or the installer before Send.

2. **Even unsent, it helps:** each question shows what we already know and what we assume if it is skipped.

3. **The homeowner picks what goes.** No map at all still gets a priced offer.

OnlyPanels



Everything we use, and what for

The home

Bavaria LoD2 3D buildings: roof sides, slope, direction, area, eave

DOP20 aerial photo, 20 cm

Bavaria single-tree map: shade per roof side

EU PVGIS 5.3: yearly sun per surface

Census 2022, 100 m grid: build year and heating nearby

Bavarian monument map

Munich heat plan: district heating

Photon: address search

Claude vision on the photo: roof windows, chimneys, old panels

The homeowner's answers, an optional meter cabinet photo

Learning and checking

Reonic, 1,041 past projects, 2 installers: sizes, battery catalog, wallbox habits

21,652 offer lines: what an offer is made of

Similar past homes: the honest range for this request

PVGIS hourly, Munich 2018–2020: own power, battery, feed-in

HTW Berlin calculator: calibrates own power

Fraunhofer ISE field tests: heat pump efficiency

276 held-out projects, split by customer: the score

Money and law

The planner's price book: parts, labour, extras

SWM 2026: power 32.4 ct, gas 11.8 ct, base fee

EEG feed-in 7.7 ct, Bundesnetzagentur

KfW BEG heat pump grant, July 2026

Munich FKG heat pump grant

Solarspitzengesetz: negative-price hours, 60% cap

MsbG smart meter fees

§14a EnWG grid fee rebate

EEG 2027 draft: payback if switched on later

When a fact is missing

Stromspiegel 2025: use by household size

128,365 Munich roofs: what a "small" or "large" roof means

Floors from eave height, living area from footprint

Gas or oil bill → heat need

Driving → car power, 0.18 kWh/km

Prices +3% a year, panels −0.4%

Every assumption is shown to the planner and must be confirmed before Send

Smart where it helps. Deterministic where it counts.

AI agents Claude, ElevenLabs

Reads the roof photo for windows, chimneys, old panels

Understands the planner: "only 18 panels fit" limits the design

Runs the desk by voice through MCP tools: open, change, ask the price, send

Guides the homeowner by voice, in English or German

Engine and rules same answer every time

Sizes solar, battery, wallbox and heat pump like this installer

Prices with grants and energy law, payback over 20 years

Asks only what moves the design: reruns itself at the edges of each guess

Owns every number. One the AI did not get from the engine is dropped



Each draft shows its steps as they run.

On 276 projects it never saw: solar off by 2.65 kWp, against 3.06 for one design for every home and 4.02 for a rule of thumb, with none of this context.

scripts/final_test.py · per installer in docs/data/findings.md

Better facts, not a bigger model.

A first design and its price in minutes. The planner stays in control. The homeowner decides what is shared.

github.com/atahankilc/onlypanels

Python · FastAPI · scikit-learn · React · MapLibre · three.js · Claude · ElevenLabs